

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I - Semester of MSc (Microbiology) Examination March 2018

Course code & Name MS 711: Microbiology

Microbiology

Date: 26/03/2018 Day: Monday Time: 10:00 AM To 10:30 AM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	20
1.	The following was the first person to observe bacteria under microscope	
	(i) Louis Pasteur	(ii) Robert Koch
	(iii) Antony Van Leewenhoek	(iv) Joseph Lister
2.	In the following microscopy, the image appears bright against dark background	
	(i) Dark field	(ii) Phase contrast
	(iii) Fluorescent	(iv) Bright field
3.	Gram's staining is an example of the following technique	
	(i) monochrome	(ii) differential
	(iii) simple	(iv) special
4.	The following is the distinguishing feature(s) of a bacterial cell	
	(i) Histone-like proteins	(ii) 18S rRNA
	(iii) 80S ribosomes	(iv) Peptidoglycan
5.	Molecular clock should run faster in organisms with a short generation time is postulated in the following hypothesis	
	(i) Generation- time effect	(ii) Punctuated equilibria
	(iii) Molecular clock	(iv) All of the above
6.	Short gaps involving upto 20-30 nucleotides are usually caused by the following	
	(i) Gene duplication	(ii) Unequal crossing over
	(iii) DNA transposition	(iv) Slipped strand mispairing
7.	The cyanobacterium that is believed to be the extant relative of the organism that gave rise to chloroplast in eukaryotes is	

	(i) <i>Prochloran</i>	(ii) <i>Tetrahymena</i>	
	(iii) <i>Synechococcus</i>	(iv) <i>Oscillatoria</i>	
8.	The following is / are most commonly used for comparing proteins of two microbes		
	(i) Reaction with antibodies	(ii) Electrophoretic mobility	
	(iii) Regulatory properties	(iv) All of the above	
9.	The following phylum of eubacteria contain ether linked fatty acids in their cytoplasmic membrane and produce water		
	(i) <i>Aquifex</i>	(ii) <i>Thermotogae</i>	
	(iii) <i>Chloroflexi</i>	(iv) <i>Chlorobi</i>	
10.	The following is an example of microcosm		
	(i) Fermenter	(ii) Incubator	
	(iii) Winogradsky column	(iv) None of the above	
11.	The following microscopy is the most suitable microscopy to determine the relative abundance of a species in a community		
	(i) Fluorescent	(ii) Light	
	(iii) SEM	(iv) AFM	
12.	During the reproduction of the following phages the coat protein is assembled while it passes through the cytoplasmic membrane of the host cell		
	(i) M13	(ii) ϕ X174	
	(iii) MS2	(iv) T7	
13.	Cell wall in Fungi is made up of		
	(i) Chitin	(ii) Starch	
	(iii) Cellulose	(iv) NAM and NAG units.	
14.	Eosin-Methylene Blue agar is an example of the following medium		
	(i) Selective	(ii) Differential media	
	(iii) Synthetic media	(iv) Both (i) & (ii)	
15.	The stage of bacterial growth wherein there is no net bacterial growth occurs is called the		
	(i) Stationary phase	(ii) Lag phase	
	(iii) Log phase	(iv) None of the above	
16.	Which of the following methods, is not used to enumerate bacterial cells?		
	(i) FACS	(ii) Pour Plate method	
	(iii) Spread plate	(iv) Spectrophotometry	
17.	The most common microscopic techniques for the study of biofilm		
	(i) Scanning Electron microscopy	(ii) Confocal Scanning Electron microscopy	
	(iii) Light microscopy	(iv) Fluorescence microscopy	
18.	Agar is obtained from		
	(i) <i>Saccharomyces cerevisiae</i> .	(ii) <i>Gelidium</i> spp.	

	(iii) <i>Gigartina stellata</i>	(iv) None of the above	
19.	Which of the following does not belong to Kingdom Fungi?		
	(i) Zoomycota	(ii) Ascomycota	
	(iii) Chytridiomycota	(iv) Rhodophyta	
20.	Thirty-six colonies grew in nutrient agar from 1.0 ml of sample withdrawn from a solution diluted to 10^{-5} in a standard plate count procedure. How many cells were in the original sample?		
	(i) 360	(ii) 3,600	
	(iii) 3,60,000	(iv) 3,600,000	

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Date: 26/03/2018 Day: Monday Time: 10:30 AM To 1:00 PM Maximum Marks: 50

Instructions:

- 1. Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q - II Answer the following questions as directed		20
Q- II (A)	. State to which class does the following organism belong and make a neat diagram illustrating the life cycle of the following organisms a. <i>Ustilago maydis</i> Or b. <i>Saccharomyces cerevisiae</i>	3
Q- II (B)	Briefly describe (Any four) a. Different kinds of specialised cells present in Cyanobacteria. b. Cardinal temperature for microbial growth. c. Differential and Selective media employed for the growth of bacteria. d. Methods for direct enumeration of bacterial cultures. e. Different phases of bacterial growth under batch culture.	12
Q- II (C)	Write short notes (any one) a. Transcriptional control in <i>Bacillus subtilis</i> biofilms b. Continuous culture c. Citing suitable examples describe the three evolutionary events that have given rise to modern day algae.	5
SECTION -II		
Q-III Answer the following questions as directed		10
Q- III (A)	Differentiate between any two of the following a. Light and Dark field Microscopy b. Microbial biofilm and mat c. Bacterial and Archaeal cell membrane.	2
Q- III (B)	Write short notes on any two of the following a. Microbial loop	4

	b. Lyophilization c. Metabolic diversity of archaea	
Q- III (C)	Attempt any two of the following a. Discuss briefly: Sample preparation for electron microscopy b. Determine the resolving power of a microscope whose objective lens has 1.25 numerical aperture and the wave length of light used is 600 nm c. Write a short note on Diversity indices used for the study of microbial diversity	4
Q-IV	Answer the following questions as directed	10
Q- IV(A)	Differentiate between any two of the following a. Evolution by horizontal gene transfer and spontaneous mutations b. Juke and Cantor' s Model and Kimura' model c. Punctuated equilibria and Phylatic gradualism	2
Q- IV(B)	Write short notes on any three of the following a. Speciation in microbes b. Evolution of mitochondria in Eukaryotic cell c. Phylogenetic classification of bacteria d. Genomic fingerprinting for the study of microbial diversity	6
Q- IV(C)	Justify any one of the following: a. Pseudogenes derived from Pol II transcripts have molecular defects and are not expressed b. Divergence of DNA sequences may not occur even after a number of substitution mutations.	2
Q-V	Answer the following questions as directed	10
Q-V(A)	Discuss briefly any two of the following a. Numerical Taxonomy b. Replication of single-stranded negative-stranded RNA viruses c. Importance of Axenic culture for the study of microbial diversity	6
Q-V(B)	Attempt any two of the following: a. Discuss briefly: VBNC and its awakening b. How can Genetic and molecular index of a community be determined? c. Discuss briefly: Application of various microscopic techniques to study interaction between two microbial cells with examples	4